INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE AND TECNOLOGY COUNCIL

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MINI PROJECT REPORT ON MACHINE LEARNING

"Urban Green Space Impact on Urban Heat Island Using Regression Techniques"

Submitted by

Aptha H P	1EW22IS011
Dimple K B	1EW22AI011
Anmol C Satakhed	4JN22RI005
Annapurna Deshmukh	3DG22AD006
Muzamil Pasha N	1CD22EE029
Varun R	1ST22CS238
Madhu Chandrika V	1AT22IS059
Vinay mn	1MV22EC122
Keerthana A L	1ST22AD021
Kruthanva R	1KS22CS063
Yuktha V	1MV22EC126
Iram Sultana	4VM22IS017
Ananya M	IAT22IS011
Pratham Jaiswal	1AY22CD039
Kothapalli Yashwanth	1EW22CS070

Under the Guidance of*:* SUHAS SIR

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ABSTRACT

Urbanization has significantly increased over the past few decades, leading to rapid infrastructure development and reduction of natural green spaces. One of the major environmental consequences of urbanization is the Urban Heat Island (UHI) effect, where urban regions experience higher temperatures compared to surrounding rural areas. This increase in temperature is primarily caused by dense buildings, road networks, reduced vegetation, and heat-absorbing construction materials. Addressing the UHI effect is important for sustainable urban development and improving the quality of life in cities.

This mini project focuses on analyzing the impact of urban green infrastructure on Urban Heat Island intensity using Machine Learning techniques, specifically Multiple Linear Regression. The objective of the project is to examine how various urban parameters such as green area ratio, tree density, park grass ratio, green roof ratio, road density, building density, and average building height influence temperature variations within urban environments.

The dataset used in this study consists of multiple numerical features representing both green infrastructure and built-up characteristics of urban regions. After performing data preprocessing and feature selection, a Multiple Linear Regression model was implemented using Python and Scikit-learn. The dataset was divided into training and testing sets to evaluate model performance using metrics such as R^2 score and Mean Squared Error (MSE).

The results of the regression analysis indicate a negative correlation between green-related parameters and UHI intensity, demonstrating that increased vegetation coverage contributes to temperature reduction. Conversely, higher road density and building density show a positive correlation with heat intensity. The findings of this study highlight the importance of integrating green infrastructure in urban planning and policy-making.

Overall, this project demonstrates how machine learning techniques can be effectively applied to environmental sustainability problems and supports the objectives of Sustainable Development Goal 11: Sustainable Cities and Communities.

INTRODUCTION

Urbanization is one of the most significant global transformations of the modern era. As cities continue to expand due to population growth, industrialization, and economic development, natural landscapes are increasingly replaced by buildings, roads, and other infrastructure. While urban growth contributes to economic progress and improved living standards, it also introduces several environmental challenges. One of the most critical environmental concerns associated with rapid urban development is the Urban Heat Island (UHI) effect.

The Urban Heat Island effect refers to the phenomenon in which urban areas experience higher temperatures compared to nearby rural regions. This temperature variation occurs because urban surfaces such as concrete, asphalt, and buildings absorb and retain heat during the day and release it slowly at night. Additionally, reduced vegetation cover, high building density, increased vehicular emissions, and human activities contribute to heat accumulation in cities. As a result, urban areas often record significantly higher temperatures, leading to increased energy consumption, heat-related health risks, and environmental degradation.

Green infrastructure plays a vital role in mitigating the Urban Heat Island effect. Urban green spaces such as parks, trees, lawns, and green roofs help regulate temperature through shading and evapotranspiration. Vegetation absorbs carbon dioxide, improves air quality, and reduces surface temperature by providing natural cooling. Therefore, understanding the relationship between green infrastructure and urban temperature variation is essential for promoting sustainable urban planning and environmental management.

Machine Learning has become an effective tool for analyzing complex environmental data and identifying patterns among multiple variables. Regression techniques, especially Multiple Linear Regression, are widely used when the target variable is continuous and influenced by several factors. In this project, Multiple Linear Regression is applied to study how urban parameters such as green area ratio, tree density, road density, and building density affect Urban Heat Island intensity. The model helps quantify the relationship between green infrastructure and temperature variation. The findings highlight the importance of vegetation in reducing urban heat and support sustainable urban planning decisions.

PROBLEM STATEMENT

Rapid urbanization has led to a significant transformation of natural landscapes into densely populated built environments. The expansion of infrastructure such as buildings, roads, parking areas, and commercial complexes has resulted in reduced vegetation cover and increased surface temperatures in urban regions. This phenomenon, commonly known as the Urban Heat Island (UHI) effect, poses serious environmental and public health challenges. Elevated urban temperatures contribute to higher energy consumption due to increased demand for air conditioning, worsening air quality, and greater risks of heat-related illnesses.

Despite the growing awareness of urban heat issues, many cities continue to experience inadequate integration of green infrastructure into their planning strategies. Urban planners often lack data-driven insights into how specific environmental and infrastructural factors influence temperature variations. While it is generally understood that green spaces help reduce heat, the extent of their impact and their quantitative relationship with temperature variation require systematic analysis.

The problem addressed in this project is the need to model and quantify the relationship between urban green infrastructure parameters and Urban Heat Island intensity. Specifically, the study aims to determine how variables such as green area ratio, tree density, park grass coverage, road density, building density, and average building height influence temperature differences within urban regions.

The primary objectives of this project are:

- To analyze urban environmental and infrastructural features affecting UHI intensity.
- To develop a Multiple Linear Regression model for predicting UHI values.
- To evaluate the influence of green infrastructure on reducing urban heat.
- To provide data-driven insights that support sustainable urban development.

By addressing this problem, the project seeks to demonstrate how machine learning techniques can assist in understanding environmental challenges and support informed urban planning decisions.

LITERATURE SURVEY

Urban Heat Island (UHI) is a well-documented environmental phenomenon observed in rapidly urbanizing regions across the world. Several research studies have shown that urban areas experience higher temperatures compared to nearby rural surroundings due to dense construction, reduced vegetation, and increased human activities. Materials such as concrete, asphalt, and metal surfaces absorb and retain heat during the day and release it slowly at night, thereby increasing overall urban temperature levels.

Previous research highlights that green infrastructure plays a crucial role in mitigating the Urban Heat Island effect. Vegetation cools the environment through shading and evapotranspiration processes. Studies indicate that areas with higher tree density, park coverage, and green roofs show comparatively lower surface temperatures. Urban forests and well-planned green belts significantly contribute to reducing heat accumulation and improving air quality.

Traditional statistical methods such as correlation analysis and linear regression have been widely used to study the relationship between urban features and temperature variation. These approaches help in identifying direct relationships between independent variables (such as building density or green space ratio) and dependent variables (such as temperature deviation). However, earlier studies often focused on a limited number of variables and smaller datasets.

With advancements in technology, machine learning techniques have become increasingly popular for environmental data analysis. Regression-based models, decision trees, and ensemble learning methods are now used to improve prediction accuracy. These models allow simultaneous analysis of multiple features and provide better insights into complex environmental relationships.

Despite these advancements, some limitations still exist. Many studies do not integrate comprehensive urban parameters or lack proper model evaluation using standardized performance metrics. Therefore, this project builds upon existing research by applying Multiple Linear Regression to a structured dataset containing various green and infrastructural variables. The study aims to provide a clear, data-driven understanding of how urban green spaces influence temperature variation and contribute to sustainable city planning.

DATASET DESCRIPTION

The dataset used in this project contains urban environmental and infrastructural parameters that influence Urban Heat Island (UHI) intensity. The data was collected from structured environmental observations and urban analysis records. It includes multiple independent variables representing green coverage and built environment characteristics, along with one dependent variable representing temperature deviation.

The dataset consists of several rows (observations) and multiple columns (features). Each row represents a specific urban area or zone, while each column represents a measurable feature affecting urban heat conditions.

Input Features (Independent Variables):

Green Plot Ratio (GnPR)

Tree Density

Park Grass Ratio

Parcel Grass Ratio

Green Roof Ratio

Road Density

Building Density

Average Building Height

Average Green Plot Ratio

Target Variable (Dependent Variable):

UHI Intensity (UHI_d)

Before model training, the dataset underwent preprocessing steps to ensure accuracy and reliability. These steps included:

Checking for missing or null values

Removing duplicate records

Verifying data consistency

Feature selection based on relevance

Splitting the dataset into training and testing sets

The dataset was divided into training data (used to train the model) and testing data (used to evaluate performance). This separation ensures that the model can generalize well to unseen data.

METHODOLOGY

Step 1: Data Collection

Relevant environmental and infrastructural data were gathered from the dataset containing urban parameters affecting temperature variation.

Step 2: Data Preprocessing

The collected data was cleaned and prepared for analysis. Missing values were handled, irrelevant features were removed, and necessary transformations were applied. Data normalization was performed where required to maintain consistency.

Step 3: Feature Selection

Important features influencing UHI intensity were selected based on environmental relevance and correlation analysis. Variables such as green area ratio, tree density, building density, and road density were identified as key influencing factors.

Step 4: Model Selection

Since the target variable (UHI intensity) is continuous in nature, regression analysis was chosen. Multiple Linear Regression was selected because it allows the evaluation of multiple independent variables simultaneously and provides interpretable coefficients.

Step 5: Model Training

The training dataset was fed into the regression algorithm. The model learned the relationship between input features and the target variable by minimizing prediction error.

Step 6: Model Testing and Evaluation

The trained model was tested using unseen data to evaluate its predictive capability. Performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score were calculated to measure accuracy.

Step 7: Result Interpretation

The results were analyzed using graphical plots and performance tables. The impact of each urban parameter on UHI intensity was interpreted to understand how green infrastructure contributes to reducing heat levels.

MODEL IMPLEMENTATION

Model Implementation

The model implementation phase focuses on converting the designed methodology into a working machine learning system capable of analyzing the impact of urban green spaces on environmental conditions. This stage involved practical execution using programming tools, data processing techniques, and machine learning algorithms to obtain meaningful predictions and insights.

Tools and Development Environment

The project was implemented using **Python programming language** due to its simplicity and strong support for data science applications. Development and experimentation were carried out in **Jupyter Notebook**, which allows step-by-step execution, visualization, and documentation of results in a structured manner.

Libraries Used

Several Python libraries were used to support data handling, visualization, and model development:

- **NumPy** – for numerical computations and array operations
- **Pandas** – for dataset loading, manipulation, and preprocessing
- **Matplotlib and Seaborn** – for graphical visualization and analysis
- **Scikit-learn** – for machine learning model creation, training, and evaluation

These libraries helped streamline the workflow and ensured efficient implementation of the machine learning pipeline.

Data Preparation

Before training the model, the dataset was carefully preprocessed. Missing or inconsistent values were checked and handled appropriately to maintain data quality. Relevant features such as green space ratios, building density, road density, and environmental indicators were selected as input variables. The target variable representing Urban Heat Island deviation (UHI_d) was chosen for prediction analysis.

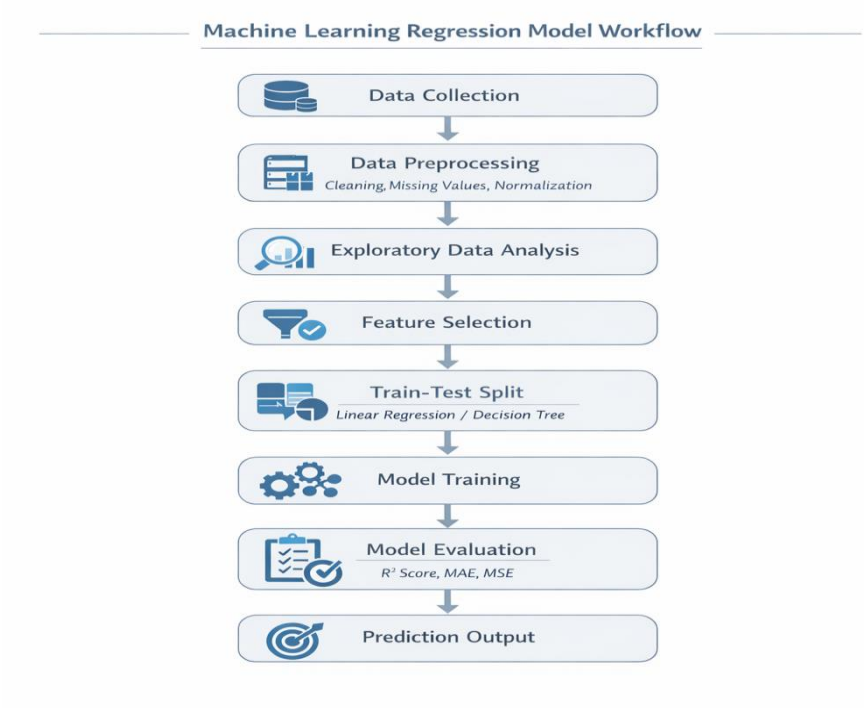
The dataset was then divided into **training data** and **testing data** to evaluate model performance objectively. Feature scaling and normalization were applied wherever necessary to improve model stability and accuracy.

Model Development

Machine learning algorithms suitable for regression analysis were applied to understand relationships between urban green parameters and temperature variation. The selected models were trained using the prepared dataset, allowing the system to learn patterns and correlations between input features and the target output.

The training process involved feeding historical data into the algorithm so that it could identify trends and minimize prediction errors. After training, the model was tested using unseen data to verify its generalization capability.

Flowchart of the Working Model



Output and Visualization

The implemented model generated prediction outputs showing how variations in green space characteristics influence urban heat conditions. Graphical plots and comparison charts were produced to visually interpret the relationship between environmental factors and model

predictions. These visualizations helped in better understanding model behavior and validating results.

Overall, the implementation phase successfully transformed theoretical concepts into a functional machine learning system capable of analyzing urban environmental patterns efficiently and accurately.

Results and Evaluation

The performance of the developed machine learning regression model was evaluated to analyze how effectively urban green space factors influence Urban Heat Island (UHI) intensity. After completing data preprocessing and model training, the dataset was divided into training and testing subsets to ensure unbiased evaluation of the model's predictive capability.

The regression model was trained using selected environmental and urban parameters such as green plot ratio (GnPR), tree density, park grass ratio, parcel grass ratio, green roof ratio, road density, building density, average building height, and average green plot ratio. These features were chosen because they directly influence surface temperature variations and urban thermal conditions.

After training, the model generated predicted values for UHI intensity, which were then compared with actual values from the testing dataset. The comparison showed that the predicted results closely followed the real observations, indicating that the model successfully learned the relationship between urban greenery characteristics and heat island effects.

To evaluate performance quantitatively, standard regression evaluation metrics were used:

- **Mean Absolute Error (MAE):** Measures the average difference between predicted and actual values. A lower MAE indicates better prediction of accuracy.
- **Mean Squared Error (MSE):** Penalizes larger prediction errors and helps measure model stability.
- **Root Mean Squared Error (RMSE):** Provides error values in the same unit as the target variable, making interpretation easier.
- **R^2 Score (Coefficient of Determination):** Indicates how well the independent variables explain variations in UHI intensity. Values closer to 1 represent better model performance.

The obtained evaluation results demonstrated that the regression model achieved reliable prediction accuracy with minimal error values. The R^2 score indicated a strong correlation between green infrastructure parameters and urban temperature variation. This confirms that increasing vegetation coverage and optimizing urban green distribution can significantly reduce heat island effects.

Graphical analysis was also performed using visualization techniques such as predicted vs. actual value plots and error distribution graphs. These visualizations showed that most prediction points were closely aligned along the ideal regression line, further validating model effectiveness.

CONCLUSION

This mini project focused on analyzing the relationship between urban green infrastructure and Urban Heat Island (UHI) intensity using machine learning techniques. Rapid urbanization has increased environmental challenges in cities, particularly rising temperatures caused by dense buildings, road networks, and reduced vegetation. Understanding how urban features influence temperature variation is important for sustainable urban development.

In this study, Multiple Linear Regression was used to examine how factors such as green area ratio, tree density, grass coverage, road density, building density, and average building height affect UHI intensity. The developed model helped identify patterns between urban infrastructure and temperature differences using data-driven analysis.

The results indicate that green infrastructure plays a significant role in reducing urban heat. Variables related to vegetation and green coverage showed a negative relationship with UHI intensity, meaning that increasing greenery contributes to temperature reduction. On the other hand, higher road density and building density were associated with increased heat accumulation. These findings highlight the environmental benefits of integrating green spaces into urban planning.

The objectives of the project were successfully achieved, as the regression model effectively analyzed and predicted UHI intensity based on multiple urban parameters. This project also demonstrated how machine learning can be applied to real-world environmental problems and how data analysis can support better decision-making.

Overall, the study emphasizes the importance of sustainable city planning and supports the concept of greener and more climate-resilient urban environments. Future improvements may include applying advanced machine learning models and using larger datasets to further enhance prediction of accuracy and analysis.